MILITARY SPECIFICATION

CARTRIDGE, 105MM, ILLUMINATING, M314A3, LOADING, ASSEMBLING AND PACKAGING

This specification is approved for use with the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department for Defense.

1. SCOPE

1.1 Scope. This specification covers Projectile, 105MM: Illuminating, M314A3 Loading, Assembling and Packaging.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-P-46565	- Primer, Percussion, M28A2 and M28B2, Loading, Assembling and Packing
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-286	- Propellants, Solid; Sampling, Examination and Testing
MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1234	- Pyrotechnics, Sampling, Inspeciton
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspections by Attributes

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US ARMY ARMAMENT RESEARCH DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

7549072 - Box, Packing, Ammo., for 105MM, How. Ammo, in Fiber Container M105A2
7549073 - Container, Ammo., Fiber, M105A2, for 105MM Howitzer Ammunition
9206744 - Projectile, 105MM: Illum., M314A3, Loading Assembly
9206821 - Cartridge, 105MM: Illum., M314A3

INSPECTION EQUIPMENT DRAWINGS

9201136 - Tunnel, Light
9201268 - Procedure, Light Output Measurement
9201390 - Procedure, Photocell Checkout
9200964 - Photocell
9200964 - Photometer Head with Cell

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquision functions should be obtained from the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E-18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS

TITLE 49 - Transportation, Parts 100-199 (The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, and referenced specifications.

3.2 Moisture content and burning time.

3.2.1 First fire composition. The moisture content of the first fire composition at the loading station, prior to loading shall not exceed 0.30 percent, when determined as specified in 4.5.2.1 or 4.5.2.3.

3.2.2 Iluminant composition. The moisture content of the illuminant composition at the loading station prior to loading, shall not exceed 0.30 percent, when determined as specified in 4.5.2.2 or 4.5.2.3.

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3.2.3 Burning time - candle power of the illuminant assembly (dwg. 9206749). The illuminant assembly during static testing shall not have a burning time of less than 77 seconds minimum (min.). The luminosity of all the illuminant assemblies shall not have an average candle power of less than 590,000 (min.).

3.3 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.5.1.

3.3.1 Functioning. The cartridge shall function satisfactorily (see 6.4) and shall provide effective illumination (see 6.10) for 55.0 seconds.

3.3.2 Premature bursts. There shall be no premature burst in the bore of the weapon or in the first 100 yards of flight.

3.3.3 Safety. There shall be no premature burst of indication of metal parts separation in the gun or in flight when fired with propellant adjusted to give 112 percent of the rated maximum pressure of the cartridge in a new tube.

3.4 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.5 Control dimensions.

3.5.1 Tool control dimensions. Dimensions marked tool control shall be gaged after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor.

3.5.2 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a .40% AQL.

3.6 Workmanship. The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.6.1 Threads. Threads shall be undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

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3.6.2 Burr and sharp edges. No part shall have a burr or sharp edge which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.6.3 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

3.6.4 Painting and marking. Painting and marking shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with applicable specification.

3.6.5 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.6.6 Protective coatings. Protective coatings shall be in accordance with the applicable drawing and the specifications listed thereon.

3.6.7 Cloth. All cloth shall be free of tears, rips and foreign matter. The stitches in the cloth shall be neat and continuous without breaks, tears or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated:

Part Description	Drawing	Quantity
Illuminating Canister (prior to Loading Illuminating Composition)	9206749	5
Illuminating Canister	9206749	5
Illuminating Canister & Parachute Assy (prior to crimping)	9206746	5
Illuminating Canister & Parachute Assy	9206746	5
Projectile Loading Assy (prior to assy of Illum. Canister & Parachute Assy)	9206744	5
Projectile Loading Assy (prior to securing base plug & closing plug)	9206744	5
Projectile Loading Assy	9206744	30
Primer (prior to Assy to cartridge case)	9206821	5
Cartridge Case, Prior to Assy to Projectile Assy	9206821	5
Cartridge Assembly	9206821	30

Thirty of the above projectile assemblies and cartridge assemblies shall be forwarded to a Government proving ground to ballistic testing. Ballistic testing will b e performed subsequent to successful completion of inspections of Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Projectile, 105MM, Illuminating, M314A3	1 OF 2	See Charts
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Illuminating Canister (Dwg. 9206749) (Prior to loading illuminating Composition) Examination for defects	5		3.1	4.4.2.1
	Illuminating Canister (Dwg. 9206749) Examination for defects	5		3.1	4.4.2.2
	First Fire Composition	4oz		3.2.1	4.4.3.4.1/4.5.2.1
	Illuminating compositon test	4oz		3.2.2	4.4.3.4.1/4.5.2.2
	Static functioning	5sets		3.2.3	4.4.3.5/4.5.3
	Illuminating Canister & Parachute Assy (Dwg. 9206746) (prior to crimping Examination for defects	5		3.1	4.4.2.3
	Illuminating Canister & Parachute Assy (Dwg. 9206746) Examination for defects	5		3.1	4.4.2.4

NOTES:

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 105MM, Illuminating, M314A3		SHEET 2 OF 2		DRAWING NUMBER: See Contents NEXT HIGHER ASSEMBLY: N/A
	Projectile Loading Assy (Dwg. 9206744) (prior to assy of Illuminating Canister & Parachute Assy) Examination for defects	5		3.1	4.4.2.5
	Projectile Loading Assy (Dwg. 9206744) (prior to securing base plug and closing plug) Examination for defects	5		3.1	4.4.2.6
	Projectile Loading Assy (Dwg. 9206744) Examination for defects Proving ground test	5 30		3.1 3.3	4.4.2.7 4.4.3.2
	Cartridge Case (Dwg. 9206821) (prior to assembling to projectile assembly) Examination for defects	5		3.1	4.4.2.9
	Cartridge Assembly (Dwg. 9206821) Examination for defects	5		3.1	4.4.2.10

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspections lots of projectiles shall contain:

a. Projectile metal parts assemblies of one lot interfix number from one manufacturer.

b. Expelling charge assemblies from one lot interfix number from one manufacturer.

c. Canister assemblies of one lot interfix number from one manufacturer.

d. Parachute assemblies from one lot interfix number from one manufacturer.

e. Primers of one lot interfix number from one manufacturer.

f. Cartridge assemblies of one lot interfix number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-1235 or MIL-STD-105 as applicable.

b. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize capability in accordance with contract.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.1	Illuminating Canister, Prior to Loading Illuminating Composition	1 1	9206749
			NEXT HIGHER ASSEMBLY 9206749

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing, damaged or improperly assembled		0.40%	3.1	Visual
102	Liner composition present on open end of canister between brakes		0.40%	3.1	Visual
Minor					
201	Coating missing or inadequate		0.65%	3.6	Visual
202	Point stake missing or inadequate		0.65%	3.6	Visual
203	Evidence of poor workmanship (other than coating)		0.65%	3.1	Visual/Manual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Illuminating Canister		9206749
			NEXT HIGHER ASSEMBLY 9206746

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Total length, maximum (max.)		0.40%	3.1	Gage
102	Any component missing, damaged or improperly assembled		0.40%	3.1	Visual
103	Disc assembly above flush		0.40%	3.1	Visual
104	First fire composition	*		3.2.1	4.4.3.4.1/4.5.2.1
105	Illuminating composition test	*		3.2.2	4.4.3.4.1/4.5.2.2
106	Static functioning	*		3.2.3	4.4.3.5/4.5.3
Minor					
201	Disc assembly coating missing or incomplete (prior to assy into canister)		0.65%	3.6	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES: * Sample size as specified in applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.3	Illuminating Canister and Parachute Assembly, Prior to Crimping	1 OF 1	9206746
			NEXT HIGHER ASSEMBLY 9206746

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Length of center wire		0.40%	3.1	Gage
Minor 201	Staking missing or improper assy of swivel pin nut		0.65%	3.1	Visual
202	Wire not frayed or improperly soldered		0.65%	3.1	Visual
203	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9206746
4.2.2.4	Illuminating Canister and Parachute Assy				NEXT HIGHER ASSEMBLY 9206744
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component or operation missing or improper		0.40%	3.1	Visual
102	Swivel fails to operate freely		0.40%	3.1	Visual
103	Staking or pettman cement missing or improper		0.40%	3.1	Visual
104	Parachute torn		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.2.2.5	Projectile Loading Assy, Prior to Assy of Illuminating Canister & Parachute Assy		1 1		9206744
					NEXT HIGHER ASSEMBLY 9206744
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component or operation missing or improper		0.40%	3.1	Visual
102	Parachute improperly folded (see dwg. 9206746, sheet 2)		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.2.2.6	Projectile Loading Assy, Prior to Securing Base Plug & Closing Plug		1	1	9206744
					NEXT HIGHER ASSEMBLY 9206744
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any component missing or improper		0.40%	3.1	Visual
102	Parachute wrapper torn		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.2.2.7	Projectile Loading Assembly				9206744
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Position of shear pin		0.40%	3.1	Gage
102	Position of twist pin		0.40%	3.1	Gage
103	Diameter of bourrelet and rear or rotating band, maximum (if retouched)		0.40%	3.1	Gage
104	Gap between base plug & base, max.		0.40%	3.1	Gage
105	Total length		0.40%	3.1	Gage
106	Shear or twist pin(s) not staked in place or missing		0.40%	3.1	Visual
107	Shear pin(s) above flush		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.7	Projectile Loading Assembly	2 OF 2	9206744
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Color of band incorrect		0.65%	3.1	Visual
202	Protective coating with bare spot in excess of 1/4 square inch		0.65%	3.6	Visual
203	Marking missing, misleading or identifiable		0.65%	3.6	Visual
204	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.2.2.8	Primer, Prior to Assy tro Cartridge Case		1 OF 1	9206821
				NEXT HIGHER ASSEMBLY 9206821

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	One or more flash holes missing		100%	3.1	
Major	None defined				
Minor 201	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.9	Cartridge Case, Prior to Assembling to Projectile Assembly	1 OF 1	9206821
			NEXT HIGHER ASSEMBLY 9206821

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propelling charge bag missing or torn		0.40%	3.1	Visual
102	Foil side of charge No. 5 not facing primer		0.40%	3.1	Visual
103	Propelling charge assembled incorrectly		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.10	Cartridge Assembly	1 OF 1	9206821
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primer above flush		100%	3.1	Gage
2	Profile and diameter of mouth		100%	3.1	Gage/4.4.3.3
Major					
101	Fuze fails to withstand the required disassembly torque		0.40%	3.1	Torque test
102	Primer loose or missing		0.40%	3.1	Visual/Manual
103	Fuze setting incorrect		0.40%	3.1	Visual
Minor					
201	Gap between fuze & projectile		0.65%	3.1	Visual
202	Fuze not staked		0.65%	3.1	Visual
203	Rtoating band damaged		0.65%	3.1	Visual
204	Marking missing, misleading or unidentafiable		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.11	Unsealed Fiber Container				7549073
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Packing component missing		0.40%	3.1	Visual
Minor 201	Cartridge cannot be removed by hand		0.65%	3.1	Manual
202	Container cap cannot be removed by hand		0.65%	3.1	Manual
203	Glue or asphalt on cartridge		0.65%	3.1	Visual
204	Packing components improperly assembled		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.12	Sealed Fiber Container	1 OF 1	7549073
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container cut or damaged through all asphalt impregnated layers		0.40%	3.1	Visual
102	Metal end loose or distorted		0.40%	3.1	Visual
103	Tape incomplete or badly wrinkled		0.40%	3.1	Visual
Minor					
201	Tear tab length, minimum		0.65%	3.1	Gage
202	Contents loose		0.65%	3.1	Manual
203	Gap between cover and body container exceeding 1/8 inch		0.65%	3.1	Manual
204	Cuts, scuffs, or gouges in outer layers		0.65%	3.1	Visual
205	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual
206	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-20354C (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7549072
4.2.2.13	Sealed Wood Packing Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged exposing contents		0.40%	3.1	Visual
102	Hardware or strapping missing, broken or loose		0.40%	3.1	Visual
103	Board broken		0.40%	3.1	Visual
Minor					
201	Strapping or hardware improperly assembled		0.65%	3.1	Visual/Manual
202	Handle missing or insecure		0.65%	3.1	Visual/Manual
203	Car seal missing, unsealed or improperly positioned		0.65%	3.1	Visual
204	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual
205	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Table and herein.

4.4.3.1 Check test for possible deterioration of Primer, Percussion, M28 (see MIL-P-46565 (AR)). If the total elapsed time between original acceptance of the M28 primer lot and the assembly of that lot into the complete round cartridges exceeds two years, or if the M28 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-46565 (AR) immediately before the M28 primer lot is assembled into the cartridges. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the complete cartridges (see 6.11).

4.4.3.2 Proving ground. The following test shall be performed as specified in 4.5.1.

4.4.3.2.1 Functioning.

4.4.3.2.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, 80 cartridges shall be selected for this test. If nine of more cartridges fail to function satisfactorily or to provide effective illumination for at least 55 seconds, the lot shall be rejected.

4.4.3.2.1.2 After the three consecutive lots have met the criteria of 4.4.3.2.1.1, 34 cartridges shall be selected for this test. If five or more fail to function satisfactorily or to provide effective illumination for at least 55 seconds the lot shall be rejected. In the event that three successive lots fail to meet the requirements of 4.4.3.2.1.2, the sampling provisions of 4.4.3.2.1.1 shall again apply. When three consecutive lots have again passed the requirements contained therein, the provisions of 4.4.3.2.1.2 shall be reinstated.

4.4.3.2.2 Premature bursts (see 3.3.2), Critical defect, Code No. 08014. If any of the sample cartridges used in any of the tests fail to comply with the requirements for premature bursts the lot shall be rejected.

4.4.3.2.3 Safety.

4.4.3.2.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, 10 cartridges shall be selected for this test. If any cartridge fails to comply with the applicable requirements the lot shall be rejected.

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4.4.3.2.3.2 After three consecutive lots have met the criteria of 4.4.3.2.3.1, three cartridges shall be selected from each lot for this test. If any cartridge fails to comply with the applicable requirements the lot shall be rejected. In the event a lot fails to meet the requirements of 4.4.3.2.3.2, the sampling provisions of 4.4.3.2.3.1 shall again apply. When three consecutive lots have again successfully passed the requirements cntained therein, the provisions of 4.4.3.2.3.2 shall be reinstated.

4.4.3.2.4 Refiring. If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional rounds as required.

4.4.3.3 Fit of projectile with cartridge case. The cartridge cases shall be gaged 100 percent for profile and diameter of the mouth to assure a proper fit with the projectile. Any case which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4. (Note: This inspection shall not be required if it is determined that the cartridge cases have been previously gaged 100 percent for these characteristics).

4.4.3.4 Moisture content.

4.4.3.4.1 First fire composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant composition sample from each eight hours of production and place it in an airtight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.4.2 Illuminant composition. The contractor shall provide adequate controls to insure compliance with the requirements and shall select a three to four ounce illuminant composition sample from each eight hours of production and place it in an airtight labeled container indicating the name of the material and date of production. If the sample, after analysis, fails to comply with the applicable requirement, the eight hours of production represented by the sample shall be rejected.

4.4.3.5 Static test of illuminant canister assembly. The following sampling plan will apply with the sample tested as specified in 4.5.3 and in accordance with Dwg. 9206749.

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4.4.3.5.1 Each inspection lot will consist of eighty five (85) candles produced consecutively by the mixing pressing operation and will be designated as one mix/batch (see 6.8) for identification purposes.

4.4.3.5.1.1 Select a sample of five (5) candles from each inspection lot (85 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 samples) comply with the specified requirements.

4.4.3.5.1.2 After successful completion of 4.4.3.5.1.1, reduce the sampling frequency to five candles from each group of six inspection lots (510 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 samples) comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.1.

4.4.3.5.1.3 After successful completion of 4.4.3.5.1.2, reduce the sampling frequency to five candles from each group of fifteen inspection lots (1275 candles). The canister assembly shall be observed for compliance with the requirements specified in 3.3. Continue this sampling frequency until 15 consecutive samples (15 groups of 5 candles) comply with the specified requirements. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.2.

4.4.3.5.1.4 After successful completion of 4.4.3.5.1.3, reduce the sampling frequency to five candles from each eight hours of production or 1615 candles, whichever is less. The canister assembly shall be observed for compliance with the requirements specified in 3.3. In the event that three consecutive lots fail to comply with the requirements, the sampling shall revert to 4.4.3.5.1.3.

4.4.3.5.1.5 In the event of any production stoppage greater than one week, the sampling plan shall revert to 4.4.3.5.1.2 and in the event of production stoppage greater than two weeks, the sampling plan shall revert to 4.4.3.5.1.1. If the continuous mixing operation has not been shut down, but merely diverted to another item, the sampling plan shall revert to 4.4.3.5.1.2.

4.4.3.5.1.6 In the event of failure of any sample to comply with the requirements, each individual inspection lot within the mix/batch must be tested. Failure of an inspection lot to comply with the requirements shall result in rejection of that particular inspection lot.

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4.4.4 Inspection equipment. For inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.5 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration checks during production. The procedures shall include as a minimum: frequency, method of verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410. As a minimum, operating personnel shall be Level I and calibrating personnel shall be Level II.

4.5 Methods of inspection.

4.5.1 Proving ground. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired with M67 propelling charges in a gun for which the ammunition is standard and observation made for conformance to all requirements (see 6.7).

4.5.1.1 Functioning. The sample cartridges shall be fired at zone 7.

4.5.1.2 Safety. The sample cartridges shall be fired with propelling charges adjusted to give 112 percent of the rated maximum pressure for the cartridge. In addition, for the first three production lots from each manufacturer (loading facility), a smear camera shall be used to observe the projectile in flight in order to record any possible malfunctioning.

4.5.2 Moisture content.

4.5.2.1 First fire composition. The moisture content of the first fire composition shall be determined in accordance with Standard MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of carbon tetrachloride and anhydrous methanol.

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4.5.2.2 Illuminant composition. The moisture content of the illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in Standard MIL-STD-1234, Method 101.2 up to paragraph 5.3.0. A 50 ml. sample of 1:1 dimethylformamidemethyl alchol solution shall be added into a 250 ml. Erlenmeyer flash (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmoshperic moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to set 5-15 minutes. A blank sample of the 1:1 dimethylformamide-methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back titrated in this solvent by adding a slight excess of Karl for at least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary and point is reached. All succeeding operations shall be completed within 3.5-0.25 minutes. Twenty-five (25) ml. of supernatant solution shall be rapidly introduced, accurately pipetted from the supernatant liquid into the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide-methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation: $\frac{AB}{5(10)}$ = _____ Percent H_2O
Less _____ Std. Blank
Leaves _____ Percent Moisture in Sample

Where: A = Mls of Karl Fischer to titrate sample
B = Standardization Factor (mg./ml.)
10 = Constant Factor
5 = Standard Sample Weight

4.5.2.3 Alternate method for first fire and illuminant composition moisture. The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4 Standard MIL-STD-286. The temperature setting shall be 100 degrees C and time setting 30 minutes; the sample size shall be between 0.40 and 0.60 grams.

4.5.3 Static test of illuminant canister assembly. The discs shall be removed from the candle prior to testing. In calculating the candle power of any one illuminant assembly, the best continuous 60 seconds of illuminometer readings are to be used.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The cartridges shall be packaged in accordance with dwg. 7549073.

5.2 Packing.

5.2.1 Level A. The containers shall be packed in accordance with dwg. 7549073.

5.3 Marking. Marking shall be in accordance with dwg. 7549072 and 7549073, as applicable.

6. NOTES

6.1 Intended use. The projectile metal parts contained in this specification are intended for use with Projectile, 105MM, Illuminating: M314A3.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, USAMCCOM (ARDEC), ATTN: AMSMC-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirement List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement Part 27, Sub-part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	Data Requirement Title	Applicable DID
3.2, 3.3	Procedure Report	DI-R-5372*
4.4.4	Inspection Equipment Description Documentation	DI-R-10054

*Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

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6.5 Submission of ballistic test data and data cards. In addition to the normal distribution, one copy of all ballistic data and ammunition cards shall be forwarded to Commander, USAMCCOM (ARDEC), ATTN: AMSMC-QAR-Q (D), Picatinny Arsenal, NJ 07806-5000.

6.6 Visual examination qualification. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.7 Static burning time. Minimum static burning time as referenced herein (see 3.8.4) is required, as a time base for the determination of luminosity and compliance does not in itself guarantee final ballistic minimum burning time requirements.

Structual integrity of the candle under set back loading is of prime importance in meeting the ballistic burning time requirement and is attained by proper formulation, blending, consolidation and cure.

6.8 Mix/batch. A mix of candles is defined as that quantity of candles which has been loaded with one mix of illuminant or all increments. A batch is defined as a quality of candles produced by a mixing process intended to make the product essentially uniform.

6.9 Functioning. "Satisfactory functioning" shall be defined as functioning of the fuze, ejection of the parachute and illuminant assembly, and proper suspension and ignition of the illuminant.

6.10 Effective illumination. Effective illumination is defined as burning of the canister without exploding or breaking up while suspended by the parachute. The time recorded for "effective illumination" shall be considered as the time from the appearance of first light from the candle to either (1) burnout of the candle while still in the air or (2) burning of the candle until ground impact. Burning of the canister after striking the ground shall not be considered as "effective illumination". Any occurrences which detract from "effective illumination" as herein defined, shall be reported in the official firing record."

6.11 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridges, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

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6.12 Subject term (keywork) listing.

Ammunition
Artillery
Illuminating
Military Specification
Munition
Projectile, 105MM: Illuminating, M314A3, Loading Assembling and Packing

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

Project (1315-A980)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-20354C (AR)	19 May 1988

3. DOCUMENT TITLE

CARTRIDGE, 105MM, ILLUMINATING, M314A3, LOADING, ASSEMBLING AND PACKAGING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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